



Real tractor beam

The Australian National University has managed to push and pull a particle through 20cm using a hollow laser beam <http://dgit.in/TractorBeams>



Future of Umbrellas

Air Umbrella is one small umbrella that shoots air into the air to protect you from getting wet <http://dgit.in/FutUmbrella>

Space Age

And the Science Gets Done

The primary objective of the Mars Orbiter Mission had been accomplished. India had successfully placed a spacecraft in Mars orbit on its maiden voyage. The satellite even began a neighbourly twitter-chat with NASA's Curiosity Rover on Mars saying "Howdy @MarsCuriosity? Keep in touch. I'll be around." But there was still more work left to do. The payload being carried by MOM consisted of five scientific instruments which would now take over and carry forward the secondary mission.

As MOM continues to make the steep elliptical orbit around Mars, with a period of almost 73 hours, it has begun to perform a range of duties. The orbit will bring the craft as low as (periapsis) nearly 422 kilometers and as far out as (apoapsis) nearly 77,000 kilometers from the surface of Mars. The five instruments loaded on board weigh in at a modest 15 kilograms but the data they would generate would be worth its weight in gold (or maybe more).

The instruments on board consist of a variety of tools for different tasks. The Lyman-Alpha Photometer on board is being used to measure the quantities of deuterium and hydrogen in the upper atmosphere of Mars for example. The data is intended to help scientists estimate the amount of water that is lost to outer space from Mars' surface. The specially designed Methane Sensor for Mars is intended to locate, map and measure any potential methane sources on the planet's surface while the Mars Exospheric Neutral Composition Analyser or MENCA is a complex bit of instrumentation used for mass spectrometry. It's a type of a quadrupole mass analyser that filters ions from the environment based

on their mass/charge ratio, which gives the neutral composition of particles in the Mars exosphere. The next bit of gadgetry onboard is the Thermal Infrared Imaging Spectrometer which takes detailed readings and measurements of the temperature and heat emissions from the Martian surface. These data sets are used to map the topographic composition of the planet as well as gather data on the mineralogy of the planet. The final but perhaps the most exciting instrument onboard Mangalyaan is the Mars Colour Camera which will take photographs of its environment and transmit them back to ISRO.

The Mars Orbiter Mission reached Mars with twice as much fuel as was expected. This could conceivably extend the lifespan of the craft to about a year orbiting and studying Mars. With its basic instrumentation, Mangalyaan will conduct atmospheric studies, environmental studies and extensive surface imaging of Mars. And with images already being received by ISRO we know that MOM has already gone above and beyond the call of duty and expectation. The only question that remains is, "What next?"

To Infinity and Beyond!

As we sit here and bask in the reflected glory of ISRO's MOM scientists, researchers and engineers who made this great achievement possible, it's heartening to know that their work has been rewarded. While ISRO has been justly rewarded with a 50 percent rise in its annual budget (to ₹6,000 crore) for 2014-15 we are assured that new and more ambitious missions to Mars are being prepped for 2018. We can only imagine what they'll come up with but

until then, we may have bigger questions to ponder ourselves.

So we can either slap each other in the back with congratulations and praises, take pride for sharing the same general geographic identity as those who actually accomplished this feat, or we can reflect. We have invested in ISRO but what about the investments we need to make for 50 years in the future? Yes, India may generate the largest number of engineers



MOM takes some great pics! A snapshot of a duststorm on Mars

but what about the scientists? The pure researchers? The Aryabhata and the Bose? Our investment in the pure sciences doesn't even come close to the ISRO budget. That perhaps is the next worthy endeavor.

Great accomplishments are not the product of luck or genius. They are a cumulative commitment made by society to its own future. These commitments come in many forms, the most significant of which is the investment we make in our educational future.

How we educate the next generation will decide if we are prepared to take the lead in international, and even interplanetary affairs. We have just stepped out into our backyard but will we be ready to trek across the stars? Are we only going to invest in a future, content with manufacturing professionals eager to move abroad? Or are we ready to start fostering the growth of discoverers motivated to take India forward? The future of our nation could depend on this decision. Until then, we can take solace in knowing that Mars doesn't demand a H1-B visa for Indians.



ISRO's Mars Orbiter
@MarsOrbiter



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@MarsOrbiter has a quirky sense of humour on Twitter.